DNA: Cracking the Code

A gene is like a long sentence which gives instructions on how to make part of a person.

Each sentence consists of lots of three letter 'words'. These words are written in a code, using only the letters A, T, C and G (abbreviations for the bases Adenine, Thymine, Cytosine and Guanine).

Each word is called a **codon**, and codes for a piece of protein called an **amino acid**.

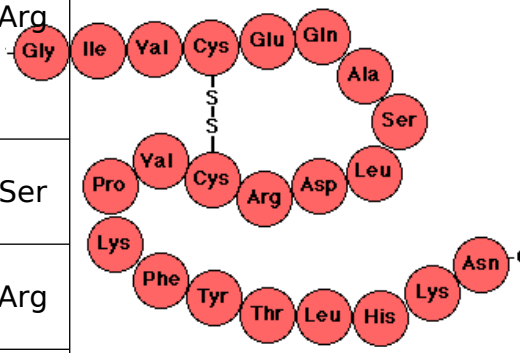
These amino acid join together in a chain to form a **protein**, which determines something about you – for example your hair colour.

The Genetic Code for Amino Acids

2nd letter→ / 1st letter	T		C		A		G	
T	TTT	Phe	TCT	Ser	TAT	Tyr	TGT	Cys
	TTC		TCC		TAC		TGC	
	TTA	Leu	TCA		TAA	Stop	TGA	Stop
	TTG		TCG		TAG		TGG	Trp
C	CTT	Leu	CCT	Pro	CAT	His	CGT	Arg
	CTC		CCC		CAC		CGC	
	CTA		CCA		CAA	Gln	CGA	
	CTG		CCG		CAG		CGG	
A	ATT	Ile	ACT	Thr	AAT	Asn	AGT	Ser
	ATC		ACC		AAC		AGC	
	ATA		ACA		AAA	Lys	AGA	Arg
	ATG	Met	ACG		AAG		AGG	
G	GTT	Val	GCT	Ala	GAT	Asp	GGT	Gly
	GTC		GCC		GAC		GGC	
	GTA		GCA		GAA	Glu	GGA	
	GTG		GCG		GAG		GGG	

TThe three letter abbreviations (Phe, Leu, etc) are the names of the amino acids.

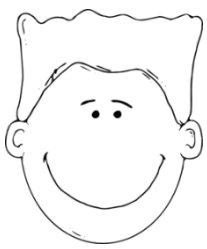
A protein is made up of a chain of amino acids



Here are the proteins that determine hair colour:

Met-Ser-Thr-Gln-Phe = Red hair
Met- Ser-Thr-His-Leu = Blonde hair
Met-Pro-Thr-His-Phe = Black hair
Met-Pro-Thr-Gln-Leu = Brown hair

Using the decoder above, find out what colour hair each of these boys will have, according to their hair colour gene. (Hint: separate each codon with a dash eg. TAG/GAT/CCA…)

DNA	ATGTCAACTCATCTCTAA	ATGCCAACTCATTTCTAG	ATGTCAACTCAATTCTAG
Protein			
Hair colour			